Latihan Transportasi dan Penugasan

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A. Masalah Transportasi (Simpleks Transportasi / MODI)

1. Data

	D_1	D_2	D_3	D_4	Pasokan
S_1	2	2	2	1	**3**
S_2	10	8	5	4	**7**
S_3	7	6	6	8	**5**
Permintaan	**4**	**3**	**4**	**4**	15

Σ pasokan = 3 + 7 + 5 = 15; Σ permintaan = 4 + 3 + 4 + 4 = 15 → seimbang.

Jumlah variabel basis = m + n − 1 = 3 + 4 − 1 = 6.

2. Solusi basis layak awal

a. Northwest Corner

Langkah	Sel	Alokasi	Sisa pasokan	Sisa permintaan
1	x_{11}	3	0, 7, 5	1, 3, 4, 4
2	x_{21}	1	0, 6, 5	0, 3, 4, 4
3	x_{22}	3	0, 3, 5	0, 0, 4, 4
4	x_{23}	3	0, 0, 5	0, 0, 1, 4
5	x_{33}	1	0, 0, 4	0, 0, 0, 4
6	x_{34}	4	0, 0, 0	0, 0, 0, 0

Z = 3(2) + 1(10) + 3(8) + 3(5) + 1(6) + 4(8) = 6 + 10 + 24 + 15 + 6 + 32 = 93

b. Least Cost

Langkah	Sel (biaya)	Alokasi	Sisa pasokan	Sisa permintaan
1	x_{14} (1)	min(3, 4) = 3	0, 7, 5	4, 3, 4, 1
2	x_{24} (4)	min(7, 1) = 1	0, 6, 5	4, 3, 4, 0
3	x_{23} (5)	min(6, 4) = 4	0, 2, 5	4, 3, 0, 0
4	x_{32} (6)	min(5, 3) = 3	0, 2, 2	4, 0, 0, 0
5	x_{31} (7)	min(2, 4) = 2	0, 2, 0	2, 0, 0, 0
6	x_{21} (10)	min(2, 2) = 2	0, 0, 0	0, 0, 0, 0

	D_1	D_2	D_3	D_4	Pasokan
S_1	2	2	2	1 **3**	**3**
S_2	10 **2**	8	5 **4**	4 **1**	**7**
S_3	7 **2**	6 **3**	6	8	**5**
Permintaan	**4**	**3**	**4**	**4**	15

$Z = 3(1) + 2(10) + 4(5) + 1(4) + 2(7) + 3(6) = 3 + 20 + 20 + 4 + 14 + 18 = 79$

c. VAM

Langkah	Penalti baris	Penalti kolom	Dipilih	Alokasi
1	1, 1, 0	5, 4, 3, 3	kolom D_1 (5)	$x_{11} = \min(3, 4) = 3$
2	–, 1, 0	3, 2, 1, 4	kolom D_4 (4)	$x_{24} = \min(7, 4) = 4$
3	–, 3, 0	3, 2, 1, –	baris S_2 (3, seri)	$x_{23} = \min(3, 4) = 3$
4	sisa baris S_3			$x_{31} = 1$, $x_{32} = 3$, $x_{33} = 1$

	D_1	D_2	D_3	D_4	Pasokan
S_1	2 **3**	2	2	1	**3**
S_2	10	8	5 **3**	4 **4**	**7**
S_3	7 **1**	6 **3**	6 **1**	8	**5**
Permintaan	**4**	**3**	**4**	**4**	15

$Z = 3(2) + 3(5) + 4(4) + 1(7) + 3(6) + 1(6) = 6 + 15 + 16 + 7 + 18 + 6 = 68$

d. RAM ($\Delta_{ij} = c_{ij} - \bar{u}_i - \bar{v}_j$)

Langkah	$\bar{u}_i$	$\bar{v}_j$	Δ terkecil	Alokasi
1	2, 10, 8	10, 8, 6, 8	−14 (x_{24})	$x_{24} = 4$
2	2, 10, 7	10, 8, 6	−11 (x_{23})	$x_{23} = 3$
3	2, –, 7	7, 6, 6	−7 (x_{11})	$x_{11} = 3$
4	7	7, 6, 6	−7 (x_{31})	$x_{31} = 1$
5	6	6, 6	−6 (x_{32})	$x_{32} = 3$
6	6	6	−6 (x_{33})	$x_{33} = 1$

Hasil sama dengan VAM, Z = 68.

Ringkasan

Metode	Z awal
Northwest Corner	93
Least Cost	79
VAM	68
RAM	68

3. Iterasi MODI (mulai dari Least Cost)

Rumus: $u_i + v_j = c_{ij}$ (basis, $u_1 = 0$); $\bar{c}_{ij} = c_{ij} - (u_i + v_j)$ (non-basis); optimal jika semua $\bar{c}_{ij} \geq 0$.

Iterasi 0 (Z = 79)

$u_1+v_4 = 1$; $u_2+v_1 = 10$; $u_2+v_3 = 5$; $u_2+v_4 = 4$; $u_3+v_1 = 7$; $u_3+v_2 = 6$

$u_1 = 0 \rightarrow v_4 = 1 \rightarrow u_2 = 3 \rightarrow v_1 = 7, v_3 = 2 \rightarrow u_3 = 0 \rightarrow v_2 = 6$

	D_1	D_2	D_3	D_4	Pasokan	u_i
S_1	2 $\bar{c} = -5$	2 $\bar{c} = -4$	2 $\bar{c} = 0$	1 **3**	**3**	0
S_2	10 **2**	8 $\bar{c} = -1$	5 **4**	4 **1**	**7**	3
S_3	7 **2**	6 **3**	6 $\bar{c} = 4$	8 $\bar{c} = 7$	**5**	0
Permintaan	**4**	**3**	**4**	**4**	15	
v_j	$v_1 = 7$	$v_2 = 6$	$v_3 = 2$	$v_4 = 1$		

$\bar{c}_{11} = 2 - (0+7) = -5$; $\bar{c}_{12} = 2 - (0+6) = -4$; $\bar{c}_{13} = 2 - (0+2) = 0$; $\bar{c}_{22} = 8 - (3+6) = -1$; $\bar{c}_{33} = 6 - (0+2) = 4$; $\bar{c}_{34} = 8 - (0+1) = 7$

Masuk basis: x_{11} ($\bar{c}$ paling negatif, −5). Loop: $x_{11}(+) \rightarrow x_{14}(-) \rightarrow x_{24}(+) \rightarrow x_{21}(-)$

	D_1	D_2	D_3	D_4	Pasokan
S_1	2 $+\theta$	2	2	1 $3-\theta$	**3**
S_2	10 $2-\theta$	8	5	4 $1+\theta$	**7**
S_3	7	6	6	8	**5**
Permintaan	**4**	**3**	**4**	**4**	15

$\theta = \min(3, 2) = 2 \rightarrow x_{21}$ keluar. Z = 79 + 2(−5) = 69

Iterasi 1 (Z = 69)

$u_1+v_1 = 2$; $u_1+v_4 = 1$; $u_2+v_3 = 5$; $u_2+v_4 = 4$; $u_3+v_1 = 7$; $u_3+v_2 = 6$

$u_1 = 0 \rightarrow v_1 = 2, v_4 = 1 \rightarrow u_2 = 3 \rightarrow v_3 = 2 \rightarrow u_3 = 5 \rightarrow v_2 = 1$

	D_1	D_2	D_3	D_4	Pasokan	u_i
S_1	**2** (2)	$\bar{c} = 1$ (2)	$\bar{c} = 0$ (2)	**1** (1)	**3**	0
S_2	$\bar{c} = 5$ (10)	$\bar{c} = 4$ (8)	**4** (5)	**3** (4)	**7**	3
S_3	**2** (7)	**3** (6)	$\bar{c} = -1$ (6)	$\bar{c} = 2$ (8)	**5**	5
Permintaan	**4**	**3**	**4**	**4**	15	
v_j	$v_1 = 2$	$v_2 = 1$	$v_3 = 2$	$v_4 = 1$		

$\bar{c}_{12} = 2 - (0+1) = 1$; $\bar{c}_{13} = 2 - (0+2) = 0$; $\bar{c}_{21} = 10 - (3+2) = 5$; $\bar{c}_{22} = 8 - (3+1) = 4$; $\bar{c}_{33} = 6 - (5+2) = -1$; $\bar{c}_{34} = 8 - (5+1) = 2$

Masuk basis: x_{33}. Loop: $x_{33}(+) \rightarrow x_{31}(-) \rightarrow x_{11}(+) \rightarrow x_{14}(-) \rightarrow x_{24}(+) \rightarrow x_{23}(-)$

	D_1	D_2	D_3	D_4	Pasokan
S_1	$2 + \theta$ (2)	(2)	(2)	$1 - \theta$ (1)	**3**
S_2	(10)	(8)	$4 - \theta$ (5)	$3 + \theta$ (4)	**7**
S_3	$2 - \theta$ (7)	(6)	$+ \theta$ (6)	(8)	**5**
Permintaan	**4**	**3**	**4**	**4**	15

$\theta = \min(2, 1, 4) = 1 \rightarrow x_{14}$ keluar. $Z = 69 + 1(-1) = 68$

Iterasi 2 (Z = 68)

$u_1+v_1 = 2$; $u_2+v_3 = 5$; $u_2+v_4 = 4$; $u_3+v_1 = 7$; $u_3+v_2 = 6$; $u_3+v_3 = 6$

$u_1 = 0 \rightarrow v_1 = 2 \rightarrow u_3 = 5 \rightarrow v_2 = 1, v_3 = 1 \rightarrow u_2 = 4 \rightarrow v_4 = 0$

	D_1	D_2	D_3	D_4	Pasokan	u_i
S_1	**3** (2)	$\bar{c} = 1$ (2)	$\bar{c} = 1$ (2)	$\bar{c} = 1$ (1)	**3**	0
S_2	$\bar{c} = 4$ (10)	$\bar{c} = 3$ (8)	**3** (5)	**4** (4)	**7**	4
S_3	**1** (7)	**3** (6)	**1** (6)	$\bar{c} = 3$ (8)	**5**	5
Permintaan	**4**	**3**	**4**	**4**	15	
v_j	$v_1 = 2$	$v_2 = 1$	$v_3 = 1$	$v_4 = 0$		

$\bar{c}_{12} = 2 - (0+1) = 1$; $\bar{c}_{13} = 2 - (0+1) = 1$; $\bar{c}_{14} = 1 - (0+0) = 1$; $\bar{c}_{21} = 10 - (4+2) = 4$; $\bar{c}_{22} = 8 - (4+1) = 3$; $\bar{c}_{34} = 8 - (5+0) = 3$

Semua $\bar{c} \geq 0 \rightarrow$ optimal.

Rekap iterasi

Iterasi	Z	Masuk ($\bar{c}$)	θ	Keluar
0	79	x_{11} (−5)	2	x_{21}
1	69	x_{33} (−1)	1	x_{14}
2	68	optimal		

4. Simpulan

Rute	Jumlah	Biaya/unit	Biaya
$S_1 \rightarrow D_1$	3	2	6
$S_2 \rightarrow D_3$	3	5	15
$S_2 \rightarrow D_4$	4	4	16
$S_3 \rightarrow D_1$	1	7	7
$S_3 \rightarrow D_2$	3	6	18
$S_3 \rightarrow D_3$	1	6	6
Total	**15**		**68**

Biaya transportasi minimum Z = 68 (solusi tunggal, tidak ada $\bar{c} = 0$).

B. Masalah Penugasan (Branch-and-Bound)

1. Data

	J_1	J_2	J_3	J_4
M_1	10	9	7	8
M_2	5	8	7	7
M_3	5	4	6	5
M_4	2	3	4	5

Model: min $Z = \Sigma\Sigma\, c_{ij}x_{ij}$; $\Sigma_j\, x_{ij} = 1$; $\Sigma_i\, x_{ij} = 1$; $x_{ij} \in \{0,1\}$. 4 mesin = 4 job → seimbang.

2. Aturan

Cabang: pilih mesin untuk J_1, lalu J_2, lalu J_3. LB simpul = biaya yang sudah ditetapkan + biaya terkecil tiap job lain (mesin boleh terpakai ganda).

Layak jika keempat mesin terpakai tepat sekali (memberi UB). Pangkas jika LB > UB. Ekspansi simpul aktif ber-LB terkecil. Selesai jika LB = UB.

3. Simpul 1 (akar)

Tiap job ke mesin termurah: $J_1 \rightarrow M_4$ (2), $J_2 \rightarrow M_4$ (3), $J_3 \rightarrow M_4$ (4), $J_4 \rightarrow M_3$ (5). LB = 2 + 3 + 4 + 5 = 14. M_4 terpakai 3 kali → tidak layak. UB = ∞.

	J_1	J_2	J_3	J_4
M_1	10	9	7	8
M_2	5	8	7	7
M_3	5	4	6	**[5]**
M_4	**[2]**	**[3]**	**[4]**	5

[ ] = pilihan biaya terkecil

4. Iterasi

Iterasi 1: ekspansi simpul 1

Simpul	Keputusan	Perhitungan LB (* = tetap)	LB	Status
2	M1→J1	10 + 3 + 4 + 5 = 22 (M1→J1* = 10; M4→J2 = 3; M4→J3 = 4; M3→J4 = 5)	22	aktif
3	M2→J1	5 + 3 + 4 + 5 = 17 (M2→J1* = 5; M4→J2 = 3; M4→J3 = 4; M3→J4 = 5)	17	aktif
4	M3→J1	5 + 3 + 4 + 5 = 17 (M3→J1* = 5; M4→J2 = 3; M4→J3 = 4; M4→J4 = 5)	17	aktif
5	M4→J1	2 + 4 + 6 + 5 = 17 (M4→J1* = 2; M3→J2 = 4; M3→J3 = 6; M3→J4 = 5)	17	aktif

LB = min(22, 17, 17, 17) = 17; UB = ∞

Iterasi 2: ekspansi simpul 5

Simpul	Keputusan	Perhitungan LB (* = tetap)	LB	Status
6	M4→J1, M1→J2	2 + 9 + 6 + 5 = 22 (M4→J1* = 2; M1→J2* = 9; M3→J3 = 6; M3→J4 = 5)	22	tak layak, dipangkas
7	M4→J1, M2→J2	2 + 8 + 6 + 5 = 21 (M4→J1* = 2; M2→J2* = 8; M3→J3 = 6; M3→J4 = 5)	21	tak layak, dipangkas
8	M4→J1, M3→J2	2 + 4 + 7 + 7 = 20 (M4→J1* = 2; M3→J2* = 4; M1→J3 = 7; M2→J4 = 7)	20	layak → UB = 20

Simpul 2 (22) ikut dipangkas. LB = 17; UB = 20

Iterasi 3: ekspansi simpul 4

Simpul	Keputusan	Perhitungan LB (* = tetap)	LB	Status
9	M3→J1, M1→J2	5 + 9 + 4 + 5 = 23 (M3→J1* = 5; M1→J2* = 9; M4→J3 = 4; M4→J4 = 5)	23	tak layak, dipangkas
10	M3→J1, M2→J2	5 + 8 + 4 + 5 = 22 (M3→J1* = 5; M2→J2* = 8; M4→J3 = 4; M4→J4 = 5)	22	tak layak, dipangkas
11	M3→J1, M4→J2	5 + 3 + 7 + 7 = 22 (M3→J1* = 5; M4→J2* = 3; M1→J3 = 7; M2→J4 = 7)	22	layak, 22 > UB, dipangkas

LB = 17; UB = 20

Iterasi 4: ekspansi simpul 3

Simpul	Keputusan	Perhitungan LB (* = tetap)	LB	Status
12	M2→J1, M1→J2	5 + 9 + 4 + 5 = 23 (M2→J1* = 5; M1→J2* = 9; M4→J3 = 4; M3→J4 = 5)	23	layak, 23 > UB, dipangkas
13	M2→J1, M3→J2	5 + 4 + 4 + 5 = 18 (M2→J1* = 5; M3→J2* = 4; M4→J3 = 4; M4→J4 = 5)	18	aktif
14	M2→J1, M4→J2	5 + 3 + 6 + 5 = 19 (M2→J1* = 5; M4→J2* = 3; M3→J3 = 6; M3→J4 = 5)	19	aktif

LB = min(18, 19) = 18; UB = 20

Iterasi 5: ekspansi simpul 13

Simpul	Keputusan	Perhitungan LB (* = tetap)	LB	Status
15	M2→J1, M3→J2, M1→J3	5 + 4 + 7 + 5 = 21 (M2→J1* = 5; M3→J2* = 4; M1→J3* = 7; M4→J4 = 5)	21	layak, 21 > UB, dipangkas
16	M2→J1, M3→J2, M4→J3	5 + 4 + 4 + 8 = 21 (M2→J1* = 5; M3→J2* = 4; M4→J3* = 4; M1→J4 = 8)	21	layak, 21 > UB, dipangkas

LB = 19; UB = 20

Iterasi 6: ekspansi simpul 14

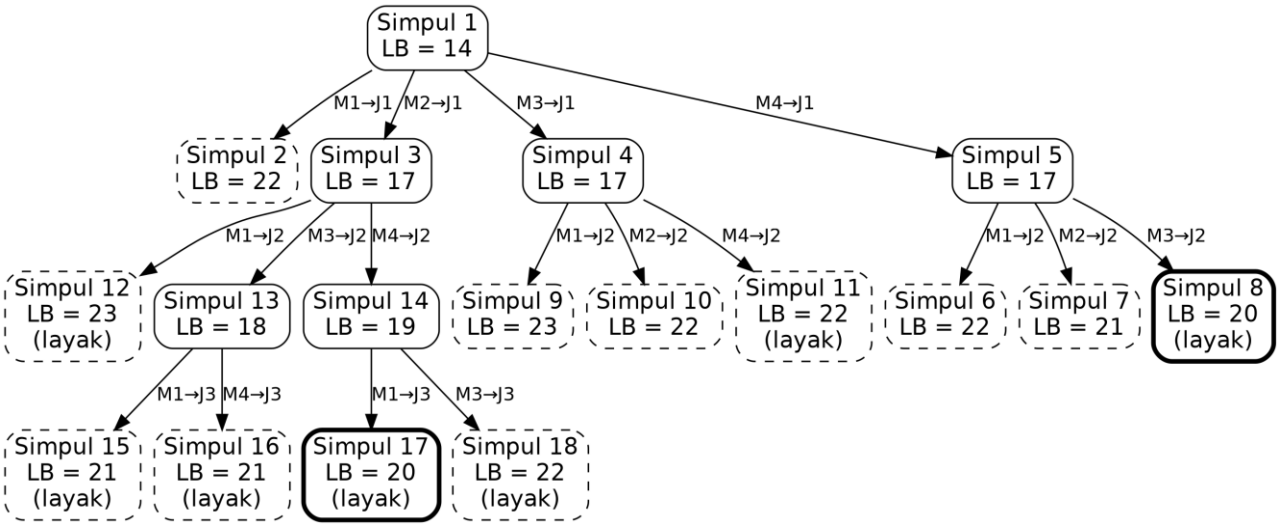
Simpul	Keputusan	Perhitungan LB (* = tetap)	LB	Status
17	M2→J1, M4→J2, M1→J3	5 + 3 + 7 + 5 = 20 (M2→J1* = 5; M4→J2* = 3; M1→J3* = 7; M3→J4 = 5)	20	layak, = UB → optimal
18	M2→J1, M4→J2, M3→J3	5 + 3 + 6 + 8 = 22 (M2→J1* = 5; M4→J2* = 3; M3→J3* = 6; M1→J4 = 8)	22	layak, 22 > UB, dipangkas

LB = 20; UB = 20 → selesai.

Rekap

Iterasi	Diekspansi	Simpul baru	LB	UB
1	1	2–5	17	∞
2	5	6–8	17	20
3	4	9–11	17	20
4	3	12–14	18	20
5	13	15–16	19	20
6	14	17–18	20	20

5. Diagram pohon



Garis tebal = optimal; garis putus-putus = layak/tak layak yang dipangkas; garis penuh = diekspansi.

6. Simpulan

	Simpul 8	Simpul 17 (alternatif)
Penugasan	$M_1 \to J_3$, $M_2 \to J_4$, $M_3 \to J_2$, $M_4 \to J_1$	$M_1 \to J_3$, $M_2 \to J_1$, $M_3 \to J_4$, $M_4 \to J_2$
Biaya	7 + 7 + 4 + 2 = 20	7 + 5 + 5 + 3 = 20

Biaya penugasan minimum Z = 20 (dua solusi optimal).